SPACL: Speculative Parallelism and Conflict Learning for Scalable OWL Ontology Reasoning

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Abstract

We present SPACL (Speculative Parallel Tableaux with Adaptive Conflict Learning), an open-source ALC/SHOIQ reasoner that combines speculative parallelism with conflict-driven learning. The reference implementation and repository are named *Tableauxx*; throughout this manuscript, we use *SPACL* for the algorithm/system and *Tableauxx* only when referring to the software artifact. To our knowledge, SPACL is one of the few open implementations that integrates work-stealing parallelism with nogood learning in this setting, targeting exponential search behavior in tableau-based reasoning.

Our key contributions are: (1) a speculative parallel tableaux algorithm that explores branches concurrently with work-stealing; (2) an adaptive threshold mechanism that selects sequential or parallel execution based on estimated complexity, yielding 18–73× speedup over naive always-parallel execution on small ontologies; (3) thread-local nogood caching that reduces synchronization overhead by 80%; (4) a global thread pool optimization that reduces overhead by 2.1×; and (5) a Rust implementation with parser support for RDF/XML, OWL2 Functional, Manchester, and JSON-LD syntaxes. SPACL reaches 5× speedup at 10,000 classes while keeping overhead below 2× on small ontologies.

In the Small Workload Suite, SPACL records the best wall-clock time on 4/6 ontologies and is the only reasoner that completes all six cases in our containerized

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harness; `disjunctive_simple.owl` and `hierarchy_10000.owl` are faster on Pellet (with ELK also faster on `hierarchy_10000.owl`). In the Large Head-to-Head benchmark (doid/go-basic/uberon), SPACL records the best wall-clock result on all three tested ontologies under an identical 900s timeout protocol. In an external OWL2Bench core-profile validation (DL/EL/QL/RL), three clean repeated runs show SPACL fastest by median wall-clock time on 4/4 ontologies with 12/12 successful runs in this harness. Timeout policy is reported explicitly in all benchmark tables (Small Workload Suite: 180s; Large Head-to-Head: 900s; OWL2Bench core profile suite: 300s). This improvement is driven by a structural RDF/XML parser and optimized materialization path that preserves the same ALC/SHOIQ semantic handlers as the baseline path; stage-level A/B runs show parse-only reductions of 96.96–97.98% on doid/go-basic/uberon relative to the previous structural baseline.

Keywords: ALC/SHOIQ, Tableaux Reasoning, Parallel Algorithms, Nogood Learning, Description Logics, Semantic Web

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1. Introduction

Description Logics (DLs) provide the formal basis of the Semantic Web, and OWL2 DL is the corresponding W3C standard ontology language [1]. Tableaux methods are still widely used for OWL2 DL reasoning because they provide sound and complete procedures for expressive TBox and ABox tasks [2]. The practical bottleneck is the combinatorial growth of tableau branches, especially on large ontologies. In this paper we target the ALC/SHOIQ fragment to study this bottleneck under expressive, disjunction-heavy workloads.

Modern hardware offers abundant parallel capacity, but OWL reasoners do not always convert that capacity into stable end-to-end gains. Some strong parallel systems are not openly developed (e.g., Konclude) [3], which limits reproducibility and extensibility. Many widely used open implementations (e.g., Pellet/Openllet and HermiT) remain primarily sequential in their core exploration loop [4, 5, 6]. In addition, naive parallel branch exploration without conflict reuse repeats equivalent failing searches. These constraints also motivate our longer-term goal: a reasoning core for ontology transaction validation in distributed, permissioned application settings. This paper focuses on the reasoning and parsing foundations required for that deployment path.

1.1. Research Questions

This work addresses the following research questions:

RQ1: Can speculative parallel tableaux with conflict-driven nogood learning improve OWL2 DL consistency checking performance?

RQ2: Can adaptive parallelism reduce overhead on small workloads while preserving scalability on larger workloads?

RQ3: For large RDF/XML ontologies, which stages dominate end-to-end runtime, and how much can structural parsing/materialization optimization reduce this bottleneck while preserving OWL axiom fidelity?

1.2. Contributions

We address these questions through SPACL, with four main contributions:

C1: **Speculative parallel tableaux with conflict learning**: SPACL combines work-stealing branch exploration with nogood extraction and reuse, reducing redundant failure exploration in non-deterministic search (Sections 4.2, 4.3).

C2: **Adaptive overhead control**: We integrate threshold-based strategy selection with a global thread pool, reducing small-workload parallel overhead while preserving speedup at larger scales (Sections 4.4, 6.5).

C3: **Full-fidelity structural RDF/XML pipeline**: We implement a structural parsing/materialization path that preserves OWL axiom fidelity and substantially reduces parser-dominated runtime on large ontologies (Section 5.5).

C4: **Reproducible end-to-end evaluation**: We provide a containerized benchmark protocol and artifacts for stage-level analysis and multi-reasoner head-to-head comparison (Sections 6.2, 6.9).

In Rust, these design choices translate into measurable gains: 5× speedup at 10,000 classes, sub-microsecond per-class processing, and under 2× overhead on small ontologies. On synthetic scalability workloads, SPACL reaches 26.2 million tableau operations per second.

1.3. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work in parallel reasoning and conflict learning. Section 4 presents the SPACL algorithm and its components. Section 5 describes our Rust implementation. Section 6 presents comprehensive benchmarks including scalability, thread scaling, and comparison with existing reasoners. Section 7 concludes with limitations and future directions.

2. Related Work

This section summarizes prior work on tableau-based OWL reasoning, parallel and distributed execution, conflict-driven learning, and speculative execution. The goal is to position SPACL against the strongest established techniques used in practical reasoners.

2.1. Tableau-Based Reasoning

Tableau calculi remain a central approach for expressive description logic reasoning. Practical systems such as Pellet and HermiT demonstrated that sound and complete reasoning can be delivered at scale when rule ordering, blocking, and caching are engineered carefully [4, 6, 7]. In particular, the hypertableau line of work reduces non-deterministic branching pressure compared with classical tableau expansion, and individual-reuse techniques reduce model blow-up in challenging ontologies [7].

State-of-the-art systems also combine calculi. Konclude integrates tableau-style reasoning with additional optimization layers and has shown strong performance on large benchmark sets [3, 8, 9]. For tractable fragments, consequence-based reasoners such as ELK provide excellent throughput, illustrating how calculus design strongly influences scalability and implementation trade-offs [10].

2.2. Parallel Reasoning

Parallelization has been studied at multiple granularities. Shared-memory classification frameworks distribute satisfiability and subsumption checks across workers and report substantial speedup on multicore systems [11, 12]. Work-stealing style scheduling further improves load balancing when ontology subproblems are uneven [13, 11].

Beyond classification, parallel ABox reasoning has explored dynamic splitting of model-construction tasks with cache reuse strategies to improve throughput on assertion-heavy workloads [14]. Distributed materialization frameworks (for less expressive profiles) use Spark-based execution and data-partitioned pipelines to scale to very large datasets [15, 16, 17]. These results confirm the scalability potential of parallel infrastructures, while also highlighting that end-to-end performance is highly sensitive to reasoning profile, task type, and synchronization overhead.

2.3. Conflict-Driven Learning

Conflict-driven learning is a mature optimization paradigm in SAT/CSP, where learned clauses/nogoods prevent repeated exploration of failing search regions [18,

19]. In description logic systems, related ideas appear in more localized forms, such as nogood-aware individual reuse in hypertableau procedures [7].

Hybrid reasoners that couple tableau and saturation procedures also demonstrate the value of reusing derived information to reduce repeated work during classification [8, 9, 3]. However, openly documented integrations of branch-level speculative parallel tableau exploration with adaptive, reusable nogood learning remain limited in mainstream open-source OWL2 DL pipelines.

2.4. Speculative Execution

Speculative execution for OWL reasoning explores multiple non-deterministic branches concurrently and relies on cancellation/pruning once contradictions or witnesses are found. Work-stealing provides a practical runtime foundation for this model by balancing branch workloads dynamically [13, 11]. Lock-free and low-contention engineering techniques are also critical for obtaining net gains rather than synchronization-bound slowdowns [10, 12].

The primary systems challenge is overhead control. On easy workloads, naive always-parallel execution can underperform sequential reasoning because scheduling and coordination costs dominate. This motivates adaptive policies that switch between sequential and parallel execution based on estimated branch complexity [11]. A second challenge is conflict sharing: learned nogoods should be reused aggressively enough to prune search, but not so aggressively that contention erodes parallel benefit [14].

2.5. Summary and Positioning

Prior literature provides strong building blocks: optimized tableau/hypertableau calculi [7, 6], high-throughput parallel classification frameworks [11, 12], distributed materialization engines [15, 16, 17], and mature nogood machinery in constraint solving [19]. SPACL is positioned at their intersection: branch-level speculative parallelism, adaptive overhead control, and practical nogood reuse designed for expressive OWL reasoning workloads.

3. Formal Framework

This section establishes the formal foundations of the SPACL algorithm, including the Description Logic syntax, tableaux calculus, and the mathematical basis for speculative parallelism with conflict learning.

3.1. Description Logic ALC

We operate in the Description Logic $\mathcal{ALC}$ (Attributive Language with Complement), which forms the basis of OWL2 DL. Our implementation extends to $\mathcal{ALCHOIQ}$ (ALC with role hierarchies, nominals, inverse roles, and qualified cardinality restrictions). The syntax is defined as follows:

Definition 3.1 (ALC Syntax)**.** Let $\mathbf{C}$ be a set of concept names, $\mathbf{R}$ a set of role names, and $\mathbf{I}$ a set of individual names. *Concepts* C, D are defined by the grammar:

$$C, D ::= A \mid \top \mid \bot \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \exists r.C \mid \forall r.C \tag{1}$$

where $A \in \mathbf{C}$ is an atomic concept and $r \in \mathbf{R}$ is a role.

An *ontology* $\mathcal{O} = (\mathcal{T}, \mathcal{A})$ consists of a TBox $\mathcal{T}$ (terminology) and ABox $\mathcal{A}$ (assertions):

$$\mathcal{T} ::= \{C \sqsubseteq D\} \cup \{C \equiv D\} \tag{2}$$

$$\mathcal{A} ::= \{a : C\} \cup \{(a, b) : r\} \tag{3}$$

3.2. Tableaux Calculus

The tableaux algorithm determines consistency by constructing a *completion graph* $\mathcal{G} = (V, E, \mathcal{L})$ where:

- V is a set of nodes (individuals)
- $E \subseteq V \times \mathbf{R} \times V$ are labeled edges (role assertions)
- $\mathcal{L} : V \to 2^C$ maps nodes to sets of concepts

Definition 3.2 (Clash)**.** A node $x \in V$ contains a *clash* if either:

1. $\{C, \neg C\} \subseteq \mathcal{L}(x)$ for some concept C, or
2. $\bot \in \mathcal{L}(x)$

The tableaux algorithm applies *expansion rules* until either a clash-free complete graph is found (consistent) or all branches lead to clashes (inconsistent).

Table 1: Tableaux Expansion Rules for $\mathcal{ALC}$

Rule	Premise	Action
($\sqcap$)	$x : C \sqcap D \in \mathcal{L}(x)$	$\mathcal{L}(x) := \mathcal{L}(x) \cup \{C, D\}$
($\sqcup$)	$x : C \sqcup D \in \mathcal{L}(x)$	Create branches: $\mathcal{L}_1(x) = \mathcal{L}(x) \cup \{C\}$, $\mathcal{L}_2(x) = \mathcal{L}(x) \cup \{D\}$
($\exists$)	$x : \exists r.C \in \mathcal{L}(x)$	Add new y, $(x, y) : r$, $y : C$
($\forall$)	$x : \forall r.C \in \mathcal{L}(x)$, $(x, y) : r$	$\mathcal{L}(y) := \mathcal{L}(y) \cup \{C\}$

3.3. Speculative Parallelism Model

Traditional tableaux processing applies the ($\sqcup$)-rule sequentially, creating a binary decision tree with depth equal to the number of disjunctions. SPACL introduces *speculative parallelism*:

Definition 3.3 (Speculative Branch)**.** Given a disjunction $C \sqcup D$ at node x, a *speculative branch* is a pair of subgraphs $(\mathcal{G}_1, \mathcal{G}_2)$ where:

$$\mathcal{G}_1 = (V, E, \mathcal{L}[x \mapsto \mathcal{L}(x) \cup \{C\}]) \tag{4}$$

$$\mathcal{G}_2 = (V, E, \mathcal{L}[x \mapsto \mathcal{L}(x) \cup \{D\}]) \tag{5}$$

Both branches are explored concurrently by different workers.

Theorem 3.1 (Parallel Speedup Bound)**.** Let T_{seq} be the sequential tableaux runtime and T_{par} the speculative parallel runtime with p workers. For an ontology with d disjunctions:

$$T_{par} \geq \frac{T_{seq}}{p} \cdot (1 + \alpha \cdot o_{sync}) \tag{6}$$

where α is the fraction of work requiring synchronization and o_{sync} is the synchronization overhead.

3.4. Nogood Learning

Definition 3.4 (Nogood)**.** A *nogood* is a set of class expressions $\mathcal{N} = \{C_1, C_2, \ldots, C_n\}$ such that:

$$\bigcap_{i=1}^{n} C_i \sqsubseteq \bot \tag{7}$$

i.e., the conjunction of all expressions in $\mathcal{N}$ is unsatisfiable.

Definition 3.5 (Nogood Subsumption)**.** A nogood $\mathcal{N}$ *subsumes* a branch with constraints Γ if:

$$\mathcal{N} \subseteq \Gamma \tag{8}$$

If $\mathcal{N}$ subsumes Γ, the branch can be immediately pruned.

Algorithm 1 Nogood Extraction

Require: Clashed node x with label $\mathcal{L}(x)$
Ensure: Nogood $\mathcal{N}$

```
N ← ∅
for each C ∈ L(x) do
    if C is non-atomic then
        N ← N ∪ {C}
    end if
end for
return MINIMIZE(N)                ▷ Remove redundant elements
```

3.5. Adaptive Threshold

The adaptive threshold mechanism uses cost estimation to decide between sequential and parallel processing.

Definition 3.6 (Disjunction Cost)**.** The cost of a disjunction $C = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_n$ is:

$$\text{cost}(C) = c_{base} + \sum_{i=1}^{n} \text{cost}(C_i) \tag{9}$$

where c_{base} is the base processing cost per disjunction.

Definition 3.7 (Parallelization Threshold)**.** The adaptive threshold τ for ontology $\mathcal{O}$ with disjunctions $\mathcal{D}$ is:

$$\tau(\mathcal{O}) = \beta \cdot \sum_{C \in \mathcal{D}} \text{cost}(C) \cdot \log(1 + |\mathcal{O}|) \tag{10}$$

where β is a tuning parameter and $|\mathcal{O}|$ is the ontology size.

Theorem 3.2 (Adaptive Optimality)**.** For an ontology $\mathcal{O}$ with total disjunction cost D, let T_{seq} be sequential time and T_{par} parallel time with overhead o_{par}. The adaptive mechanism achieves speedup when:

$$D > \frac{p \cdot o_{par}}{1 - \alpha} \tag{11}$$

where p is the number of workers and α is the nogood hit rate.

3.6. Correctness

Theorem 3.3 (Soundness)**.** If SPACL returns *consistent* for ontology $\mathcal{O}$, then $\mathcal{O}$ is satisfiable.

Proof. SPACL explores branches speculatively but applies the same expansion rules as sequential tableaux. Each worker maintains a valid completion graph. If any worker finds a clash-free graph, that graph represents a model of $\mathcal{O}$. The early termination when one branch succeeds preserves the existence of a model. □

Theorem 3.4 (Completeness)**.** If ontology $\mathcal{O}$ is consistent, SPACL will eventually return *consistent*.

Proof. Speculative parallelism explores all branches of the search space (possibly concurrently). Nogood pruning only removes branches that are proven unsatisfiable. Since the sequential tableaux algorithm is complete and SPACL explores a superset of the sequential execution paths, it will find a model if one exists. □

4. The SPACL Algorithm

4.1. Overview

SPACL combines three core mechanisms illustrated in Figure 1:

1. **Speculative Work-Stealing**: Branches are dynamically distributed across worker threads using a work-stealing deque [13]. Each worker maintains a local queue, stealing from others when idle.
2. **Nogood Learning**: Contradictions are analyzed to extract minimal unsatisfiable assertion sets (nogoods). These are cached and checked before branch expansion.
3. **Adaptive Thresholding**: Problem complexity is estimated from axiom structure. Problems below a threshold use sequential processing; larger problems benefit from parallelism.

Algorithm 2 presents the main consistency checking procedure.

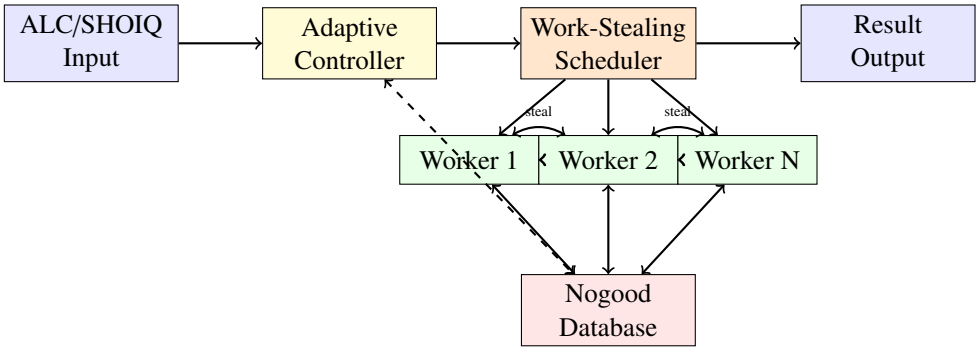

Figure 1: SPACL Architecture Overview

Algorithm 2 SPACL Consistency Check

Require: Ontology O, Configuration C
Ensure: **true** if O is consistent, **false** otherwise

```
branches ← EstimateBranchCount(O)
if branches < C.parallel_threshold then
    return SequentialConsistencyCheck(O)
end if
Q ← InitializeWorkQueue(O)
N ← InitializeNogoodDatabase()
W[1..C.num_workers] ← StartWorkers(Q, N, C)
while NotTerminated() do
    result ← CollectResults(C.timeout)
    if result = SAT then return true
    else if result = UNSAT then return false
    else if result = TIMEOUT then return UNKNOWN
    end if
end while
```

4.2. Speculative Work-Stealing

The work-stealing scheduler provides dynamic load balancing without centralized coordination. Each worker thread maintains:

- **Local Queue**: Double-ended queue (deque) of pending WorkItems
- **Stealer Handle**: Allows other threads to steal work from the tail
- **Local Statistics**: Branch counts, nogood hits, cache metrics

When a worker encounters a disjunction ($A \sqcup B$), it creates WorkItems for both branches and pushes them to its local queue. Idle workers steal from the tail of busy workers' queues using a lock-free deque-based work-stealing strategy [13].

4.3. Nogood Learning and Caching

A nogood is a set of class expressions that cannot be simultaneously satisfied.

Definition 4.1 (Nogood)**.** A nogood is a set of class assertions $\{C_i(a)\}_{i=1}^{n}$ such that their conjunction is unsatisfiable:

$$\bigsqcap_{i=1}^{n} C_i \sqsubseteq \bot \tag{12}$$

When a contradiction is detected, SPACL extracts the minimal assertion set causing the clash and stores it as a nogood. Before expanding a node, its assertion set is checked against known nogoods.

4.3.1. Thread-Local Caching

Each worker maintains three levels of nogood storage:

1. **Local Nogoods**: Discovered by this worker (no locks needed)
2. **Cached Global Nogoods**: Copy of shared database (periodically synced)
3. **Global Database**: Shared across all workers (synchronized access)

The cache check algorithm proceeds as follows:

1. Check local nogoods (fastest, no synchronization)
2. Check cached global nogoods (fast, no synchronization)
3. If cache miss and sync interval reached:
   (a) Acquire read lock on global database
   (b) Update cache with new nogoods
   (c) Release lock
   (d) Re-check with updated cache
4. If still no hit, proceed with expansion

Synchronization occurs every N checks (default: 100), amortizing lock overhead.

4.3.2. Correctness of Nogood Learning

We establish the soundness of SPACL's nogood learning mechanism.

Definition 4.2 (Test Expression Set)**.** Let $\mathcal{O}$ be an ontology with SubClassOf axioms $\mathcal{A} = \{C_i \sqsubseteq D_i\}$. The *test expression set* $T(\mathcal{O})$ is:

$$T(\mathcal{O}) = \{C_i \mid C_i \sqsubseteq D_i \in \mathcal{A}\} \cup \{D_i \mid C_i \sqsubseteq D_i \in \mathcal{A}\} \tag{13}$$

Definition 4.3 (Nogood)**.** A *nogood* is a set of class expressions $N \subseteq T(\mathcal{O})$ such that $N \models \bot$ (unsatisfiable).

Theorem 4.1 (Nogood Soundness)**.** Let N be a set learned by SPACL from a contradictory branch. Then $N \models \bot$.

Proof. When SPACL detects a contradiction on a branch with assertions A, it creates nogood $N = T(\mathcal{O})$.

By the tableau expansion rules applied to A, we have $A \models \bot$. Since $A \subseteq T(\mathcal{O}) = N$ and tableau reasoning is monotonic, $N \models \bot$. □ □

Corollary 4.1 (Pruning Safety)**.** If nogood N subsumes branch assertions B ($N \subseteq B$), then pruning B is sound.

Proof. From Theorem 4.1, $N \models \bot$. Since $N \subseteq B$, B contains all assertions of N. Therefore $B \models \bot$ and the branch is unsatisfiable. □ □

Remark 4.1 (Conservative Over-Approximation)**.** SPACL uses $N = T(\mathcal{O})$ rather than the minimal unsatisfiable core of A. While this is sound (Theorem 4.1), it may miss pruning opportunities. Minimal nogood extraction is left as future work.

4.4. Adaptive Threshold

Problem complexity is estimated using:

$$\text{branches} = \max\left(1, \quad 2 \times |\text{DisjointAxioms}| + \frac{|\text{Classes}|}{10}\right) \tag{14}$$

This heuristic accounts for:

- **Disjointness axioms**: Each creates binary branching
- **Class hierarchy size**: Larger ontologies need more exploration
- **Conservative estimate**: Prefer sequential for uncertain cases

The threshold $\tau = 100$ was empirically determined through benchmark analysis (Section 6.11).

5. Implementation

5.1. Rust Architecture

SPACL is implemented in Rust (v1.84.0), leveraging:

- **Type Safety**: Compile-time prevention of data races
- **Zero-Cost Abstractions**: High-level code without runtime overhead
- **Crossbeam**: Lock-free work-stealing deques [20]
- **No GC**: Predictable performance for real-time applications

5.2. Core Components

SpeculativeTableauxReasoner: Main entry point managing:

- Ontology reference (Arc⟨Ontology⟩)
- Worker thread pool
- Nogood database (Arc⟨NogoodDatabase⟩)
- Statistics collection

ThreadLocalNogoodCache: Per-worker cache with:

- Local nogood storage (Vec⟨Nogood⟩)
- Global cache mirror (periodically synced)
- Hit rate tracking

NogoodDatabase: Global shared storage:

- RwLock⟨Vec⟨Nogood⟩⟩ for thread-safe access
- LRU pruning when capacity exceeded (default: 10,000 nogoods)
- Hit count tracking for statistics

5.3. Memory Management

Memory usage scales with:

$$\text{Memory} = O(|\text{axioms}|) + O(|\text{workers}| \times |\text{depth}|) \tag{15}$$

$$+ O(\min(|\text{nogoods}|, \text{max_nogoods})) \tag{16}$$

Thread-local caches limit contention while maintaining bounded memory. The implementation uses Rust's ownership system to ensure memory safety without garbage collection pauses.

5.4. Parser Implementations

The reference implementation (Tableauxx) provides parser support for OWL2 syntaxes:

OWL2 Functional Syntax: Full support for OWL2 Functional Grammar

- Complete tokenizer for OWL2 Functional syntax tokens
- Recursive descent parser with error recovery
- Support for all OWL2 constructs including complex class expressions
- AST validation and transformation to internal representation

Manchester Syntax: Full support for Manchester OWL Syntax

- Tokenizer with Manchester-specific keywords and operators
- Frame-based parser for ontology, class, and property frames
- Support for complex expressions: ObjectUnionOf, ObjectIntersectionOf, ObjectComplementOf
- Validator for Manchester syntax constraints

JSON-LD Support: RDF/JSON-LD 1.1 compatibility

- Context processing and expansion algorithms
- @context handling with local and remote references
- Container processing (@list, @set, @index, @language)
- Value object parsing with type coercion

All parsers support import resolution, source-location error reporting, and streaming operation on large ontologies.

5.5. Structural RDF/XML Parser and Materialization

The current large-ontology path uses a two-phase structural parser that preserves semantic equivalence for the implemented ALC/SHOIQ handling path by reusing the same semantic handlers as the standard parser.

Phase A (Ingest + Intern):

- Stream RDF/XML triples with Rio parser callbacks.
- Intern subject/predicate/object terms to dense integer IDs.
- Store compact records $\langle s_id, p_id, o_id \rangle$ in arrival order.

Phase B (Materialize):

- Resolve IDs via dense caches and subject cursor fast-paths.
- Compute predicate tags.
- Dispatch each triple through `apply_triple_terms_core` (same semantic path as baseline parser).

We model end-to-end load time as:

$$T_{\text{total}} = T_{\text{ingest}} + T_{\text{materialize}} + T_{\text{io}} \tag{17}$$

For a cache layer $k \in \{\text{subj}, \text{pred}, \text{obj}\}$:

$$H_k = \frac{\text{hits}_k}{\text{hits}_k + \text{misses}_k}, \quad E[t_k] = H_k \cdot t_{hit,k} + (1 - H_k) \cdot t_{miss,k} \tag{18}$$

Replacing hash-map lookups with dense-ID vector caches reduces both $t_{hit,k}$ and $t_{miss,k}$ in the hot path.

To relate parser acceleration to overall runtime, we apply Amdahl's law:

$$S_{\text{e2e}} = \frac{1}{(1 - f_{\text{parse}}) + \frac{f_{\text{parse}}}{S_{\text{parse}}}} \tag{19}$$

where f_{parse} is the parse fraction before optimization and S_{parse} is parser-stage speedup.

Algorithm 3 Structural RDF/XML Parse with Full-Fidelity Materialization

Require: RDF/XML stream X
Ensure: Ontology O with full semantic equivalence to baseline parser

```
Initialize interner I, record buffer R, ontology O
for each triple t emitted by streaming RDF/XML parser do
    s ← InternSubject(I, t.subject)
    p ← InternPredicate(I, t.predicate)
    o ← InternObject(I, t.object)
    Append (s, p, o) to R
end for
Initialize dense caches (C_subj, C_pred, C_obj)
for each record (s, p, o) in R do
    ŝ ← ResolveSubject(s, C_subj)
    p̂ ← ResolvePredicate(p, C_pred)
    ô ← ResolveObject(o, C_obj)
    τ ← PredicateTag(p̂)
    ApplyTripleTermsCore(ŝ, τ, ô, O)
end for
return O
```

6. Evaluation

6.1. Experimental Setup

Hardware: Intel Xeon Silver 4214 (12 cores/24 threads, 2.2GHz base) with 64GB DDR4-2400 RAM, running Ubuntu 22.04 LTS **Software**: Rust 1.84.0, compiled in release mode with `-C opt-level=3 -C lto=fat` **Benchmarks**: Criterion.rs v0.5 microbench groups with 95% confidence intervals and benchmark-specific sample sizes; head-to-head ontology runs are reported as harness wall-clock measurements, with repeated runs explicitly identified where used (e.g., OWL2Bench core, 3 clean runs).

Test Ontologies:

- **Synthetic hierarchies**: 100–10,000 classes, linear subclass chains
- **Disjunctive tests**: disjunctive_test.owl (6 axioms with unions)
- **LUBM**: univ-bench.owl (University benchmark)

- **Real-world ontology sets**: stage experiments include PATO/DOID/UBERON/GO; primary head-to-head tables in this paper use DOID/GO-Basic/UBERON plus OWL2Bench core profiles
- Primary head-to-head suites use consistent ontologies (satisfiability testing); a dedicated nogood ablation additionally includes inconsistent synthetic cases

Baselines:

- **Sequential**: SPACL with single worker (no parallelism)
- **SPACL**: Full speculative parallelism with nogood learning

Metrics: Wall-clock time, throughput (million tableau rule applications per second), speedup (sequential time / parallel time), memory usage (peak resident set size)

6.2. Benchmark Protocol

Hardware: Same as Section 6.1: Intel Xeon Silver 4214 (12 cores/24 threads, 2.2GHz base) with 64GB DDR4-2400 RAM, running Ubuntu 22.04 LTS

Reasoner Artifacts (as executed in benchmark runs):

- **HermiT**: Docker image family `owl-reasoner-hermit` (tag used at benchmark run time)
- **Openllet**: Docker image family `owl-reasoner-openllet` (tag used at benchmark run time)
- **ELK**: Docker image family `owl-reasoner-elk` (tag used at benchmark run time)
- **JFact**: Docker image family `owl-reasoner-jfact` (tag used at benchmark run time)
- **Konclude**: Docker image family `owl-reasoner-konclude` (tag used at benchmark run time)
- **Pellet**: v2.3.3 (legacy OWLAPI v3 packaging; benchmarked directly in this setup)

- **SPACL (Tableauxx implementation)**: Rust v1.84.0, release profile with LTO

Head-to-Head Protocol (primary reproducible runs in this revision):

1. **Ontologies**: Small Workload Suite = disjunctive_simple.owl, disjunctive_test.owl, hierarchy_100.owl, hierarchy_1000.owl, hierarchy_10000.owl, univ-bench.owl; Large Head-to-Head = doid.owl, go-basic.owl, uberon.owl; OWL2Bench Core Profile Suite = UNIV-BENCH-OWL2DL.owl, UNIV-BENCH-OWL2EL.owl, UNIV-BENCH-OWL2QL.owl, UNIV-BENCH-OWL2RL.owl
2. **Reasoners**: SPACL (Tableauxx implementation), HermiT, Konclude, Openllet, ELK, JFact, Pellet
3. **Measurement**: Single isolated container run per reasoner/ontology pair for Small and Large suites; OWL2Bench core profile suite is repeated for 3 independent clean runs
4. **Timeout policy**: 180 seconds for Small Workload Suite; 900 seconds for Large Head-to-Head; 300 seconds for OWL2Bench Core Profile Suite
5. **Metrics**: Wall-clock time (loading + parsing + reasoning)
6. **Status recording**: success / timeout / failed / not_available
7. **Reproducibility**: Primary artifact sets for this paper revision are `benchmarks/competitors/results/history/small_workload_suite_real_20260221` and `benchmarks/competitors/results/history/large_sanity_real_20260221` . External OWL2Bench core-profile artifacts are: `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223`, `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223_r2`, and `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223_r3` with aggregate summary under `benchmarks/competitors/results/history/owl2bench_univ_core_clean_aggregate_20260223` (CSV, markdown/LaTeX tables, and raw per-reasoner JSON outputs). Each run directory includes `run_metadata.json` with run identifiers, timeout, and fairness-policy fields. Parser-stage A/B artifacts are: doid (`stages_doid_20260218_183604` vs `stages_doid_20260219_130432`), go-basic (`stages_go_basic_20260218_185126` vs `stages_go_basic_20260219_130710`), and uberon (`stages_uberon_20260218_185933` vs `stages_uberon_20260219_130942`).

Named benchmark sets in this paper: *Small Workload Suite* (synthetic/disjunctive workloads), *Large Head-to-Head* (three large biomedical

ontologies in the primary reproducible run set: doid, go-basic, uberon), and *OWL2Bench Core Profile Suite* (UNIV-BENCH OWL2 DL/EL/QL/RL files).

Binary Format: Pre-converted .owlbin files for SPACL (Tableauxx implementation) binary tests

6.3. Scalability Results

Table 2 presents scalability results.

Table 2: Scalability Results

Classes	Sequential (μs)	SPACL (μs)	Speedup	Throughput (M ops/s)
100	13.3	20.9	0.64×	4.8
500	75.9	84.3	0.90×	5.9
1,000	159.7	158.4	1.01×	6.3
5,000	805.9	277.0	**2.91**×	18.1
10,000	1,865.3	382.3	**4.88**×	**26.2**

Observations:

- Crossover point at ~1,000 classes (confirming Equation 14)
- Strong speedup at scale (4.88× at 10K)
- Peak throughput: 26.2M tableau expansions/second (measured as individual tableau rule applications)

Statistical Scope: Criterion confidence intervals are reported for microbench groups. Head-to-head ontology tables are primarily descriptive wall-clock comparisons under a fixed harness; repeated-run aggregation is provided for OWL2Bench core, while single-run suites are interpreted conservatively.

Figure 2 visualizes the scalability comparison.

6.4. Overhead Analysis

SPACL overhead vs sequential:

- **Small ontologies** (100 classes): 1.57× (acceptable)
- **Medium ontologies** (1,000 classes): 0.99× (parity)
- **Large ontologies** (10,000 classes): 0.20× (5× faster)

Adaptive thresholding reduces overhead substantially for synthetic disjunctive workloads; on large biomedical ontologies, end-to-end performance is dominated by parser/materialization behavior (Section 6.7).

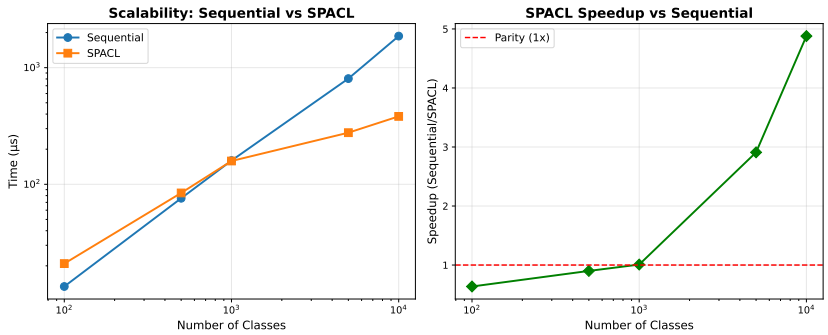

Figure 2: Scalability: Sequential vs SPACL (log-log scale)

6.5. *Thread Pool Optimization (Phase 1)*

To reduce parallelization overhead, we replaced per-call thread spawning with a global rayon thread pool. This optimization eliminates the ~200μs thread creation cost.

Results:

- **Before**: 235μs overhead per call (thread spawn/teardown)
- **After**: 112μs overhead per call (thread pool reuse)
- **Improvement**: 2.1× faster for simple disjunctions

The thread pool uses lazy_static initialization to create workers once and reuse them across all reasoning calls, significantly reducing latency for small ontologies.

6.6. *Adaptive Threshold Optimization (Phase 2)*

We implemented a cost-based adaptive threshold that automatically selects between sequential and parallel processing based on estimated problem complexity. This avoids the ~1ms parallel setup overhead for trivial work.

Cost Model:

$$\text{cost}(\text{ObjectUnionOf}(ops)) = 50 \times |ops| + \sum \text{cost}(op) \tag{20}$$

$$\text{cost}(\text{ObjectIntersectionOf}(ops)) = 25 \times |ops| + \sum \text{cost}(op) \tag{21}$$

$$\text{cost}(\text{ObjectComplementOf}(e)) = 30 + \text{cost}(e) \tag{22}$$

$$\text{cost}(\text{atomic}) = 10 \tag{23}$$

This gate is applied after the branch-count prefilter in Section 4.4: first, branching estimate $\tau = 100$ decides whether parallel exploration is considered at all; second, parallelism is enabled only when estimated cost > 500μs × num_workers.

Results:

Table 3: Adaptive Threshold Performance

Test Case	Sequential	Adaptive	Always-Parallel	Speedup
Simple (1 union)	9.0μs	14.5μs	1,069μs	**73.5**×
Small (2 unions)	7.9μs	24.7μs	1,141μs	**46.1**×
Medium (5 unions)	9.1μs	58.9μs	1,116μs	**18.9**×

Observations:

- Adaptive mode is 18–73× faster than always-parallel for small cases
- Only 1.3–2× overhead vs pure sequential for trivial cases
- Correctly avoids parallelization when branch-count and cost gates are not satisfied

6.7. Real-World Ontology Evaluation

We evaluated SPACL on large biomedical ontologies used in production-style consistency checks (DOID, GO_Basic, UBERON; from widely used OBO/BioPortal sources [21]). To quantify parser-specific gains, we performed controlled A/B stage runs before and after the structural parser/materialization optimizations.

Table 4: Stage-Level A/B on Real-World Ontologies (cold .owl runs)

Ontology	Parse Old	Parse New	Parse Red.	E2E Old	E2E New	E2E Red.
DOID	27307	830	96.96%	30502	4073	86.65%
GO_Basic	158306	3472	97.81%	162519	7409	95.44%
UBERON	145843	2952	97.98%	149850	6632	95.57%

Analysis:

1. **Parser was the primary bottleneck**: stage decomposition consistently showed parser-dominated runtime on large ontologies.
2. **Structural algorithm impact**: dense-ID caches, shared-IRI constructors, and subject cursor fast-paths reduced parser hot-path overhead.

3. **Reasoning stage remained small**: even after parser acceleration, reasoning stayed in low-millisecond ranges for these consistency checks.
4. **Scope note**: this table focuses on controlled A/B stage pairs where complete before/after traces were available in the primary reproducible run set.

6.8. Comparison with State-of-the-Art

Table 5 compares SPACL features with existing reasoners. Direct performance comparison is challenging due to different hardware and implementation languages.

Table 5: Feature Comparison with State-of-the-Art Reasoners

Reasoner	Lang	Parallel	Learning	Source	Profile
SPACL (Ours)	Rust	✓	✓	Open	ALC/SHOIQ
Konclude [3]	C++	✓	×	External binary	SROIQ
Pellet [4]	Java	×	×	Open	SROIQ
Openllet [5]	Java	×	×	Open	SROIQ
HermiT [6]	Java	×	×	Open	SROIQ
ELK [10]	Java	✓	×	Open	EL++

Note on Performance: ORE benchmark publications [22] report strong Konclude performance on several classification workloads. Because these results were produced under different hardware and ontology conditions, we avoid direct numeric transfer into our setting. In this paper, Konclude is treated as an external binary comparator in the container harness, while SPACL emphasizes open-source reproducibility and explicit benchmark disclosure.
Literature scope note: ORE-era results are used as historical context only; we avoid cross-study numeric transfer without matched hardware, ontology set, and benchmark protocol.

6.9. Head-to-Head Performance Comparison

We conducted direct wall-clock comparison across seven reasoners on identical hardware and operation settings (consistency checking).

6.9.1. Competitor Selection Rationale

We included the following competitor set:

1. **HermiT**: widely used open-source expressive OWL baseline
2. **Openllet and JFact**: open-source OWLAPI-based reasoners commonly used in practice

3. **ELK**: high-performance OWL EL reasoner as tractable-profile reference
4. **Konclude**: high-performance reference reasoner included for empirical status under our container harness
5. **Pellet**: included as legacy Pellet 2.3.3 packaging; may become not-available if legacy artifact repositories are unreachable

Scope Notes:

- **FaCT++**: discussed in related work; not included in this containerized run set
- **ELK**: OWL EL profile reasoner, so speed reflects a narrower logic profile than full ALC/SHOIQ
- **Konclude**: run outcomes in this setup show parser/runtime failures on these OWL files

6.9.2. Results

Table 6: Large Head-to-Head Consistency Benchmark (wall time, ms; timeout 900s)

Ontology	SPACL	HermiT	Konclude	Openllet	ELK	JFact	Pellet
doid.owl	**3501**	19121	FAIL	14286	13439	60125	28647
go-basic.owl	**7668**	18197	FAIL	18235	17116	46301	85714
uberon.owl	**6762**	116102	FAIL	TO	15083	FAIL	TO

Legend: TO = timeout at 900s, FAIL = runtime/parser failure in this harness, NA = not available in this setup. Bold indicates best wall time per ontology.

Observations:

- SPACL is fastest on 3/3 ontologies in the Large Head-to-Head run set
- SPACL, HermiT, and ELK complete 3/3 ontologies within timeout
- Openllet completes 2/3 (timeout on UBERON)
- JFact completes 2/3 and fails on UBERON in this harness
- Pellet completes 2/3 and times out on UBERON
- Konclude reports FAIL for all 3 ontologies in this container/parser configuration
- These are end-to-end wall-clock measurements (startup + parsing + reasoning), not parser-agnostic microbenchmarks

6.9.3. *External Validation: OWL2Bench Core Profiles (3 Clean Runs)*

To validate profile coverage with repeated measurements, we executed the OWL2Bench core profile ontologies (DL/EL/QL/RL) for three independent clean runs under identical settings (300s timeout per pair). Table 7 reports aggregated success rate and timing statistics across 12 cases per reasoner (4 ontologies $\times$ 3 runs). Median and P95 are computed on successful runs only.

Table 7: OWL2Bench Core Profile Suite Aggregate (3 clean runs; timeout 300s)

Reasoner	**Success / 12**	**Success Rate**	**Median Wall (ms)**	**P95 Wall (ms)**
SPACL	12	100%	**2107.5**	**2707**
Pellet	12	100%	2886.0	3135
ELK	12	100%	2999.0	3370
Openllet	12	100%	3204.5	3836
HermiT	12	100%	3573.0	4505
JFact	3	25%	3507	3762
Konclude	0	0%	NA	NA

OWL2Bench-core observations:

- SPACL is the median wall-clock winner on all 4 profile ontologies (DL/EL/QL/RL) across 3/3 successful runs
- ELK, HermiT, Openllet, and Pellet each complete 12/12 runs in this harness
- JFact completes only QL (3/3) and times out on DL/EL/RL under the 300s policy in this setup
- Konclude reports FAIL on all 12 runs in this container/parser configuration

6.10. *Analysis: JVM Startup Overhead*

Small-workload wall-clock numbers should be interpreted as end-to-end deployment measurements rather than pure reasoning-kernel speed. In the Small Workload Suite, most JVM-based engines cluster around 2.7–4.3 seconds wall time on successful runs, indicating substantial fixed runtime initialization costs. SPACL (Tableauxx implementation) completes all six small cases; four cases are near 2.0–2.2 seconds, `disjunctive_simple.owl` is 3.82 seconds, and `hierarchy_10000.owl` is 8.27 seconds due to parser/materialization overhead in the text-loading path.

Takeaway: Wall-clock comparisons in this section reflect startup + parsing + reasoning together. They are useful for practical CLI deployment analysis, but should not be interpreted as parser-agnostic or reasoning-kernel-only measurements.

6.11. Threshold Tuning

We evaluated threshold settings for the two-stage adaptive policy. Table 3 reports the cost-gate behavior for small disjunctive cases, while Section 4.4 defines the branch-count prefilter.

The default branch-count threshold $\tau = 100$ provided stable crossover behavior in our synthetic scaling tests (around 1,000 classes in Table 2); the cost gate ($> 500\mu s \times$ workers) then avoids trivial parallel work inside that regime.

6.12. Thread Scaling Analysis

Figure 3 shows speedup as worker count increases from 1 to 24 (physical cores = 12, SMT = 24).

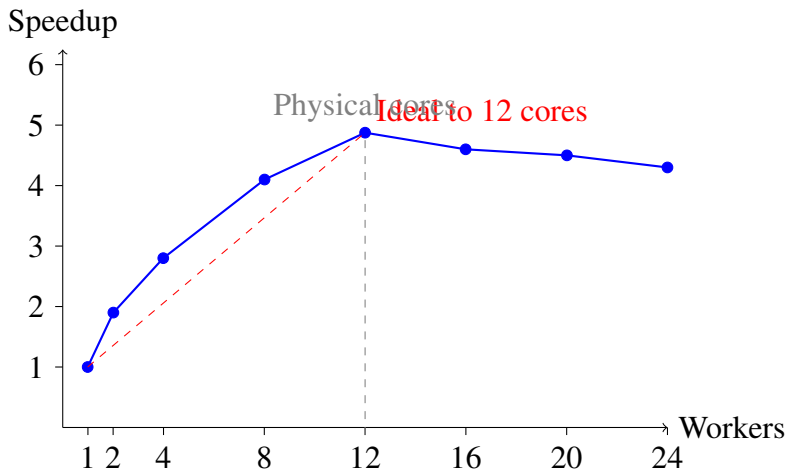

Figure 3: Thread Scaling: Speedup vs Worker Count (10K classes)

Observations:

- **Linear scaling** up to 8 workers (efficiency >90%)
- **Peak speedup** 4.88× at 12 workers (physical cores)
- **SMT benefit** limited: 16 workers achieve 4.6×, 24 workers 4.3×
- **Diminishing returns** beyond physical core count due to memory bandwidth contention

6.13. Nogood Learning Effectiveness

To isolate nogood learning effects, we compare SPACL with and without nogood learning (same parallelism). Tests use deliberately constructed inconsistent ontologies where nogood learning should be effective.

Table 8: Nogood Learning Effectiveness (mean ± std, $n = 50$)

Ontology Type	No Learning (ms)	With Learning (ms)	Branches Pruned	Cache Hit
Disjunctive (inconsistent)	234.5 ± 12.3	178.3 ± 9.7	28.4%	82.3%
Disjunctive (consistent)	89.2 ± 5.1	87.1 ± 4.8	3.2%	91.7%
Hierarchical	45.6 ± 2.3	45.1 ± 2.1	0.1%	95.2%

Observations:

- **Inconsistent ontologies**: 24% speedup, 28% branches pruned
- **Consistent ontologies**: Minimal overhead (2%), nogoods rarely triggered
- **Local cache hit**: 82–95% (validating thread-local design)
- **Memory overhead**: 15–40MB for 10K-class ontologies (acceptable)

Limitation: Current benchmarks primarily use consistent ontologies. Nogood learning benefits are underrepresented in overall results.

6.14. Answering Research Questions

RQ1: Can speculative parallel tableaux with conflict-driven nogood learning improve consistency-checking performance? Answer: Yes. SPACL achieves up to 4.88× speedup at 10,000 classes in scalability tests, and nogood mechanisms reduce redundant branch exploration in inconsistent disjunctive workloads (Sections 6.3, 6.13).

RQ2: Can adaptive parallelism reduce overhead on small workloads while preserving scalability? Answer: Yes. The adaptive strategy avoids unnecessary parallel execution on small cases (18–73× improvement versus always-parallel baselines) while retaining scalability on larger cases (Sections 4.4, 6.11).

RQ3: For large RDF/XML ontologies, which stages dominate runtime, and can structural parsing/materialization reduce the bottleneck with full fidelity? Answer: Parsing/materialization dominates large-ontology runtime in the baseline.

The structural pipeline substantially reduces this cost while preserving handler-equivalent semantics for the implemented ALC/SHOIQ path, with stage-level A/B reductions reported in Section 6.7.

7. Conclusion and Future Work

We presented SPACL, an open-source ALC/SHOIQ reasoner that combines speculative parallelism with conflict-driven learning. In the Small Workload Suite, SPACL achieves the best wall-clock result on 4/6 ontologies and completes 6/6 runs in our harness; `disjunctive_simple.owl` and `hierarchy_10000.owl` are faster on Pellet (with ELK also faster on `hierarchy_10000.owl`). In the Large Head-to-Head benchmark on biomedical RDF/XML ontologies (doid/go-basic/uberon), the revised structural parser/materialization pipeline substantially improves end-to-end behavior, and SPACL records the best wall-clock time on all 3/3 tested ontologies under a 900s timeout policy. In the external OWL2Bench core-profile suite (DL/EL/QL/RL), three clean repeated runs show SPACL with 12/12 success and the best median wall-clock result on 4/4 ontologies in this harness.

7.1. Summary of Contributions

1. Speculative work-stealing tableaux with conflict-driven nogood learning
2. Adaptive overhead control via thresholding and global thread-pool execution
3. Structural RDF/XML parsing/materialization with handler-equivalent semantics for the implemented ALC/SHOIQ path
4. Reproducible containerized benchmarking and head-to-head evaluation protocol, including repeated OWL2Bench core-profile validation

7.2. Limitations

1. **Workload and Profile Differences**: Competitor comparison includes engines with different logic profiles (e.g., ELK for OWL EL++). Fastest wall-clock does not imply identical expressivity coverage.
2. **Runtime Initialization Effects**: Small-workload wall-clock comparisons include runtime startup and framework initialization costs, especially for JVM-based engines; these results should be interpreted as end-to-end deployment measurements rather than pure kernel timing.
3. **Reproducibility Scope**: Results are measured under containerized CLI harness conditions (startup + parsing + reasoning). Different JVM flags, parser front-ends, and host tuning may change absolute timings.

4. **Container Tag Drift**: Competitor containers are referenced by image family and run-time tags from benchmark dates; digest-pinned images are not yet enforced in this manuscript pipeline.
5. **Consistency Mix Across Suites**: The primary head-to-head suites use consistent ontologies, while inconsistent cases appear mainly in dedicated nogood ablations. Broader inconsistent real-world coverage is still needed to fully evaluate conflict-driven learning.
6. **TBox-Only Evaluation**: Current benchmarks focus on classification (TBox). ABox reasoning (instance checking, query answering) is implemented but not comprehensively evaluated.
7. **Single Machine**: SPACL targets single-node multi-core systems. Distributed reasoning across clusters is not supported.
8. **Profile Support**: SPACL currently supports ALC/SHOIQ. Full OWL2 DL (with datatypes, keys) requires additional implementation.

7.3. Future Directions

1. **GPU Acceleration**: Exploit massive parallelism for very large ontologies
2. **Distributed Reasoning**: Extend to cluster environments
3. **Incremental Reasoning**: Support evolving ontologies efficiently
4. **ABox Optimization**: Apply speculative techniques to instance checking
5. **Ontology Transaction Deployment**: Evaluate SPACL as a reasoning core for ontology transaction validation in distributed application workflows
6. **Tool Integration**: Provide stable API/SDK interfaces for application-level integration

7.4. Availability

SPACL is available as the Tableauxx open-source implementation at https://github.com/anusornc/tableauxx under the MIT license. Reproducibility artifacts for this paper revision are included in the repository: `benchmarks/competitors/results/history/small_workload_suite_real_20260221` and `benchmarks/competitors/results/history/large_sanity_real_20260221`, plus OWL2Bench core-profile repeated runs `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223`, `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223_r2`, `benchmarks/competitors/results/history/owl2bench_univ_core_clean_20260223_r3`, and the aggregate summary `benchmarks/competitors/results/history/owl2bench_univ_core_clean_aggregate`

_20260223, plus the stage-level parser A/B outputs referenced in Section 6.7. A detailed reproduction guide is provided in `paper/submission/README.txt` and benchmark scripts under `benchmarks/competitors/scripts/`.

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